Veeranna N Desai

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Profile

Python Full Stack Developer skilled in FastAPI, Django REST Framework, React.js, and ML data pipelines. Delivered production-grade REST APIs and a JupedSim-based synthetic data pipeline at Trinity Mobility's AI/ML team. Looking for a full-time role where I can own backend systems end to end.

Experience

Trinity Mobility | *Python Developer Intern — AI/ML Team* *Feb 2026 – May 2026*

- Built a JupedSim synthetic data pipeline generating 10,000+ agent-level records across 20+ crowd scenarios — bottlenecks, evacuation, and surge — cutting real-world data dependency by ∼80%.
- Designed FastAPI REST endpoints for real-time, parameterized scenario generation achieving sub-200ms latency, with seamless ML Composer frontend integration.
- Delivered a 15+ variable dynamic parameter system (crowd density, agent speed, exit capacity, event triggers), enabling non-technical users to generate hundreds of scenarios and accelerating dataset creation by 3×.
- Owned the end-to-end Crowd Surge Prediction pipeline — scenario design, simulation, validation, and API delivery — across transit, stadium, and evacuation environments to improve model generalization.

Projects

SaaS Uptime Monitoring System | Python, FastAPI, PostgreSQL, React.js, JWT, Docker ↗
`github.com/veerannaNdesai/Saas-Monitoring-Platform`

- Built a full-stack uptime monitoring platform tracking availability, response status, and response time across 50+ endpoints with configurable health-check intervals (1–30 min) and a real-time React.js dashboard showing UP/DOWN/PENDING status.
- Implemented JWT authentication, secure multi-user monitor ownership, PostgreSQL-backed historical check records, and a Docker dev environment — architected to handle thousands of health-check executions per day.

Stock Price Prediction Portal | Python, Django REST Framework, React.js, LSTM, SQLite ↗
`github.com/veerannaNdesai/stock-predicton-portal`

- Built a full-stack forecasting platform with a Django REST Framework API, React.js frontend, and live Yahoo Finance data ingestion, backed by a multi-step LSTM time-series model.
- Evaluated model performance on MAE and RMSE metrics and architected a RESTful API layer decoupling ML inference from the frontend, enabling independent scaling of training and serving.

Crowd Surge Prediction System | Python, FastAPI, JupedSim, Angular, REST APIs ↗
`github.com/veerannaNdesai/Crowd-Surge-Prediction-System`

- Generated 10–15 labeled synthetic crowd-movement datasets via JupedSim simulations, closing a critical training-data gap for the ML Composer Crowd Surge Prediction model.
- Built a FastAPI backend that exposes simulation output as structured REST API endpoints, consumed by the Angular frontend and downstream ML training jobs.

Technical Skills

Backend: FastAPI, Django REST Framework, REST APIs, PostgreSQL, SQLite
Frontend: React.js, Angular, TypeScript, JavaScript (ES6+), HTML5, CSS3
ML, Data & Analytics: LSTM/RNN, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Testing & Collaboration: Pytest, Postman, Docker, Jira, Git, GitHub

Education

KLE Institute of Technology, Hubli *2021 – 2025*
Bachelor of Engineering, Electronics & Communication (ECE)